

Model Training Report

1. Models & Training Configurations

- Logistic Regression: A linear model trained with L2 regularization, max_iter=1000.
- SVM (Support Vector Machine): Linear kernel SVM with probability estimates enabled, max_iter=5000.
- Random Forest: 100 trees with default settings and random_state=42.

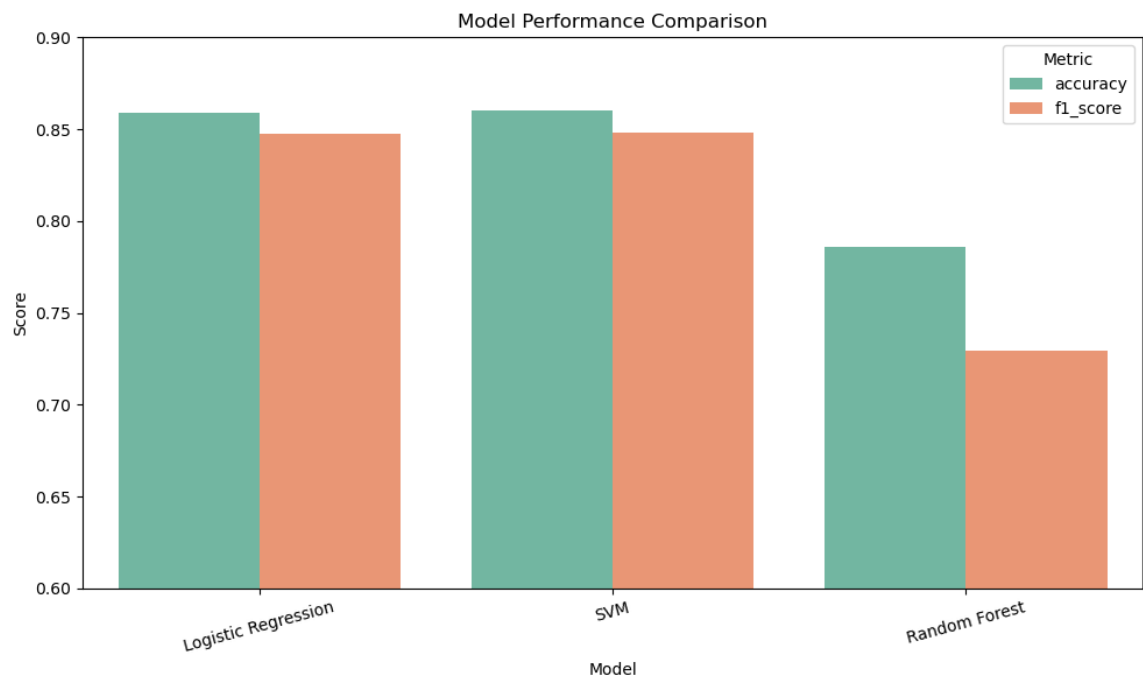
Model	Key Hyperparameters
Logistic Regression	<code>`max_iter=1000`, `solver='lbfgs`, `penalty='l2`</code>
Support Vector Machine (SVM)	<code>`kernel='linear`, `probability=True`</code>
Random Forest	<code>`n_estimators=100`, `max_features='sqrt`, `random_state=42`</code>

2. Performance Comparison

Model	Accuracy	F1 Score (Weighted)
Logistic Regression	0.8590	0.8476
SVM	0.8600	0.8481
Random Forest	0.7860	0.7295

3. Visual Performance Overview

Accuracy & F1 Score Comparison



Observation: SVM slightly outperforms Logistic Regression with the highest F1 Score and Accuracy. Random Forest performs notably worse, likely due to its behavior with high-dimensional sparse data.

Key Observations

- SVM achieved the highest accuracy (0.8600) and weighted F1 score (0.8481).
- Logistic Regression is a close second, with nearly identical performance but faster training time.
- Random Forest performed the worst, particularly struggling with minority class prediction.
- SVM appears most promising for further tuning or deployment, provided training time can be optimized.

4. Additional Insights

Classification Report Snapshots:

a. Logistic regression

Classification Report:				
	precision	recall	f1-score	support
0	0.78	0.61	0.69	3257
1	0.72	0.24	0.36	503
2	0.88	0.96	0.92	11222
accuracy			0.86	14982
macro avg	0.79	0.60	0.66	14982
weighted avg	0.85	0.86	0.85	14982

b. SVM

Classification Report:				
	precision	recall	f1-score	support
0	0.76	0.66	0.70	3257
1	0.72	0.15	0.25	503
2	0.89	0.95	0.92	11222
accuracy			0.86	14982
macro avg	0.79	0.59	0.62	14982
weighted avg	0.85	0.86	0.85	14982

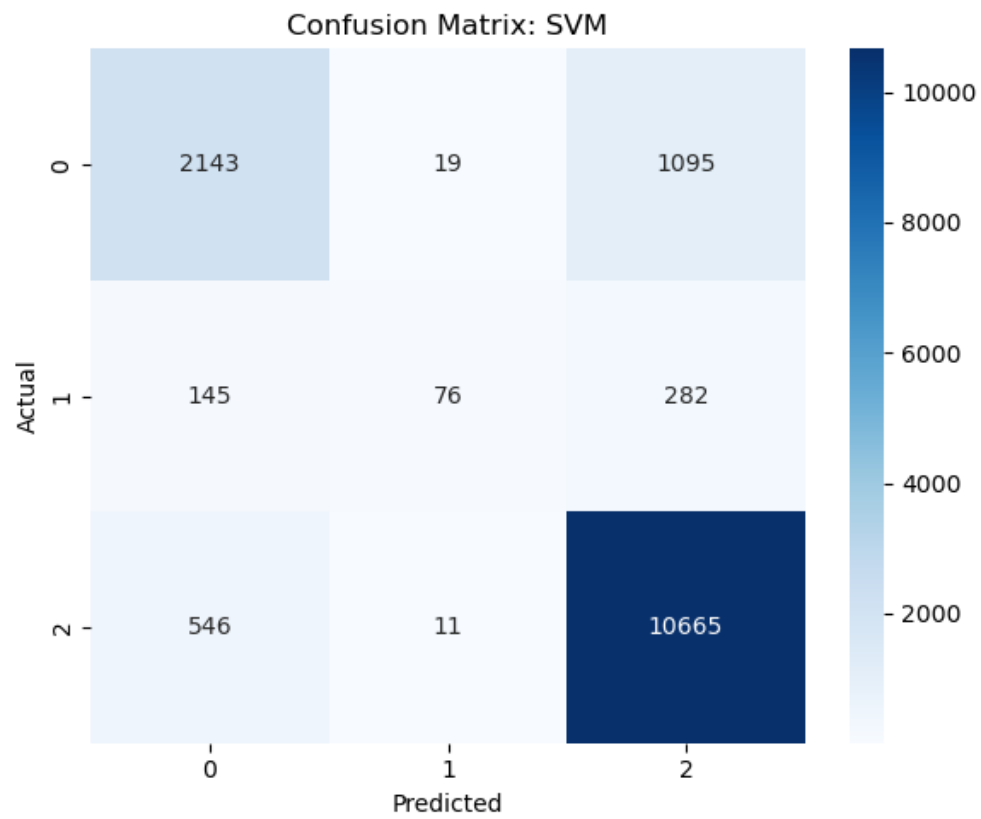
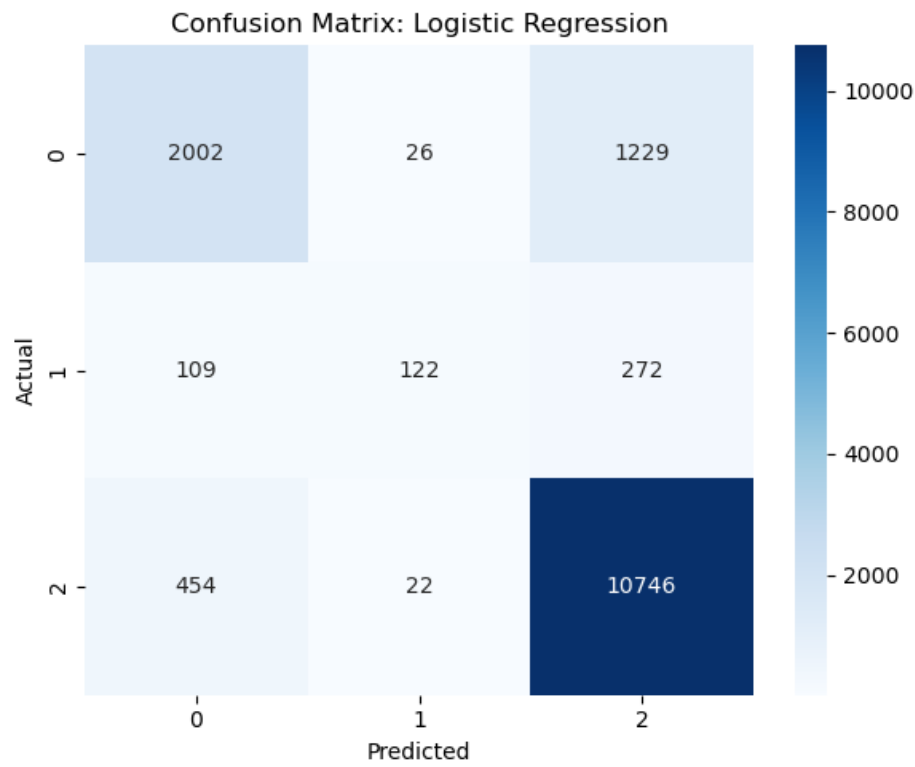
c. Random Forest

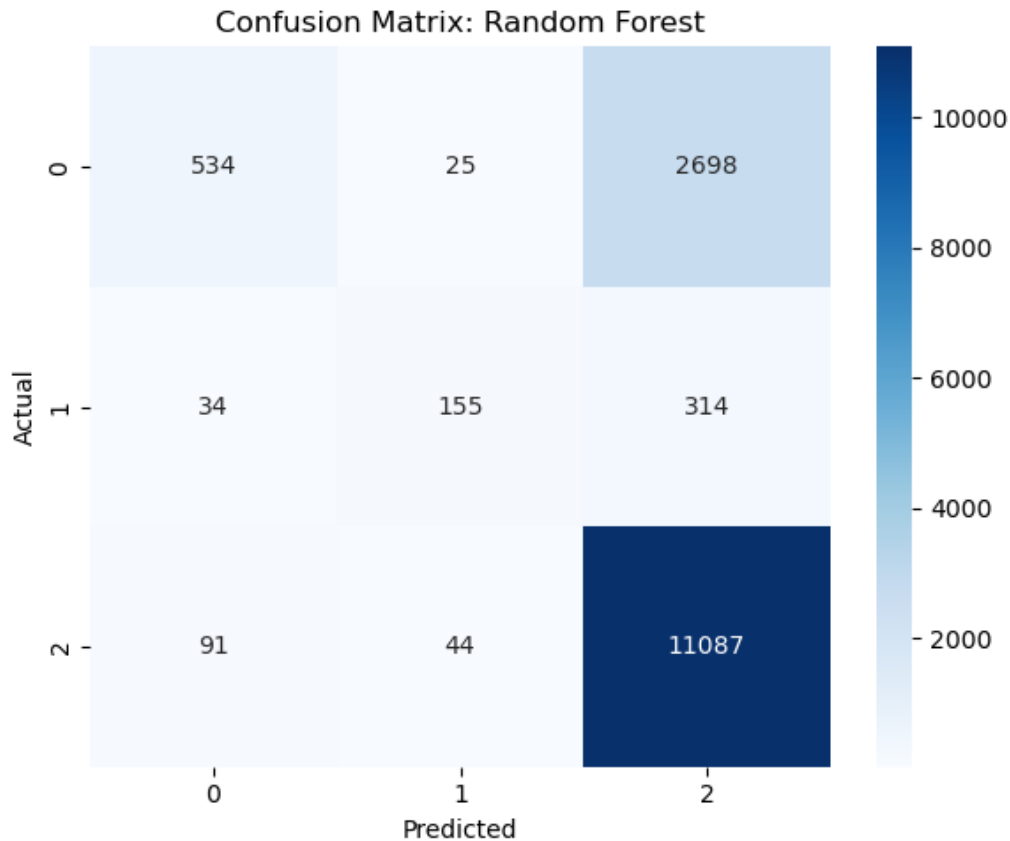
Classification Report:				
	precision	recall	f1-score	support
0	0.76	0.66	0.70	3257
1	0.72	0.15	0.25	503
2	0.89	0.95	0.92	11222
accuracy			0.86	14982
macro avg	0.79	0.59	0.62	14982
weighted avg	0.85	0.86	0.85	14982

SVM and Logistic Regression achieved high recall for class `2` but struggled on the minority class `1`.

Random Forest had low recall on class `0` and `1`, indicating poor handling of class imbalance.

Confusion Matrices:





5. Interpretation & Recommendation

- SVM slightly outperforms the other models in both accuracy (0.8600) and F1 Score (0.8481), making it the most promising candidate.
- Logistic Regression performs comparably well, offering nearly the same accuracy with less training time and complexity, making it a strong baseline.
- Random Forest, while still acceptable, lags behind in both accuracy and F1 score. However, it may still be useful if interpretability or feature importance is needed.

Recommendation: Proceed with SVM for further tuning and validation. Alternatively, use Logistic Regression for faster deployment if resources are limited.

6. Conclusion

- SVM demonstrated the best performance with a balanced F1 Score and Accuracy, making it the most promising candidate for deployment or further hyperparameter tuning.
- Logistic Regression was competitive and may serve as a fast and efficient fallback model.
- Random Forest was less effective for this sparse text data and is not recommended for further refinement.

7. Artifacts Overview

Artifact	Path
TF-IDF Features File	features_tfidf.pkl
Trained Logistic Regression	log_model_v1.joblib
Trained SVM Model	svm_model_v1.joblib
Trained Random Forest Model	rf_model_v1.joblib
Label encoder	label_encoder_v1.joblib